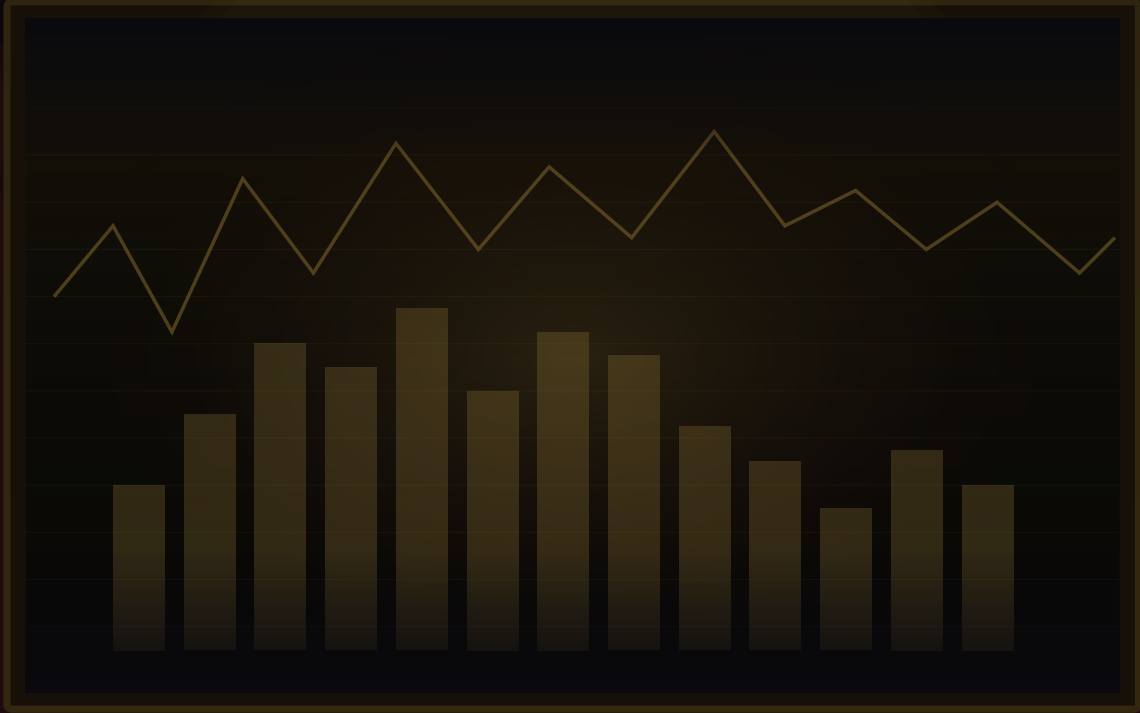


# DATAFLIX



A Deep Dive into the Data of What Gets Made,  
What Gets Seen,  
and What Gets Paid

A film directed visualized by  
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A COM480 Production

# INTRODUCTION

We all watch movies. We quote them, argue about them, revisit them on rainy Sundays. But behind every film we've ever loved sits one of the most complex, data-rich industries in the world : one that moves billions of dollars, shifts with culture, and reinvents itself decade after decade.

We wanted to understand it. Not just feel it, but actually see it.

So we collected data spanning nearly four decades of cinema (genres, budgets, revenues, ratings, production volumes) and asked ourselves the questions that don't have obvious answers: Has quality gone up or down as production exploded? Do bigger budgets actually win? Which genres survived, and which quietly disappeared?

DataFlix is our attempt to answer them visually.

Through a series of interactive visualizations, we invite you to explore how the movie industry has transformed from 1988 to 2025, to spot the long-term shifts, the sudden disruptions, and the subtle patterns hiding beneath the surface of raw data.

This project was built for anyone who has ever sat in a dark room and wondered: what is cinema, really, and where is it going?

# DATA AND PRE-PROCESSING

Our dataset came from TMDB, over 1.1 million movie entries spanning nearly a century of cinema. Rich in scope, but far from clean.

The first reality check was the missing data. Financial fields were the worst offenders: 93.5% of budget values and 97.7% of revenues were recorded as zero (not truly zero, but simply unknown). The same applied to runtime, vote counts, and ratings. Rather than letting these zeros silently distort our analysis, we replaced them with null values and treated them as missing.

Columns like genres, cast, and production companies were stored as raw comma-separated strings. We converted them into proper lists, allowing us to treat each genre or collaborator individually and track their evolution over time.

For runtime, we identified and removed statistical outliers using the IQR (interquartile range) method (keeping the dataset grounded in reality while preserving edge cases like experimental cinema where relevant).

As a result, we got a leaner and more honest dataset.

# WEBSITE AND TOOLS

## Our motivation

The design of our website was strongly inspired by interactive storytelling platforms and editorial-style data visualization websites that guide the user through a narrative experience rather than presenting isolated charts.

We wanted the exploration of cinema history to feel immersive and fluid, which is why we chose a continuous vertical scrolling structure instead of separating the content into distinct pages. This scrolling experience creates a chronological journey through the evolution of cinema, with the timeline acting as a *fil directeur* connecting all sections and visualizations together.

We were also inspired by the aesthetic evolution of cinema itself: the beginning of the website uses darker tones and black-and-white visuals reminiscent of early films, before progressively transitioning toward brighter colors and more dynamic elements as the narrative reaches modern cinema. This visual transformation allows users not only to observe the evolution of the film industry through data, but also to experience it visually and emotionally.

## How we reached it?

To implement this design, we relied on modern web technologies including HTML, CSS, and JavaScript, while using D3.js to create interactive and animated visualizations.

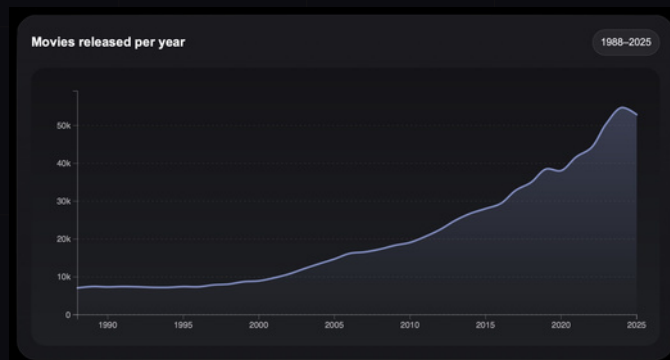
CSS Flexbox and Grid were particularly useful for structuring the scrolling layout and ensuring responsiveness across devices, while JavaScript handled dynamic interactions such as filtering, hovering, and timeline updates.

D3.js gave us the flexibility to design custom visualizations that integrate naturally into the storytelling flow of the website, rather than appearing as separate static charts.

We also used GitHub Pages for deployment and JSON files for lightweight data management, allowing the website to remain fully interactive without requiring a backend server.

# VISUALISATIONS

## Movie production over time



Even though it is one of the simplest visualizations on the website, visualization of the production over time plays an important role as the user's first entry point into the data.

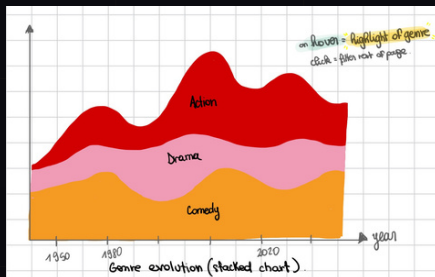
Indeed, starting with a simple and intuitive visualization allows users to immediately understand the interface without feeling overwhelmed by more complex graphics.

The direct interaction through the genre buttons makes the visualization feel dynamic and engaging from the start, encouraging users to explore the rest of the website while already immersing them in the cinematic narrative and visual atmosphere of the project.

It shows the number of movies produced each year, making it easy to observe how film production has evolved over time. We added genre selection buttons so users can filter the graph and focus on specific types of movies.

These choices are not isolated: the selected genres also influence other parts of the website, creating a more connected and interactive experience across the whole narrative.

## Genre Evolution



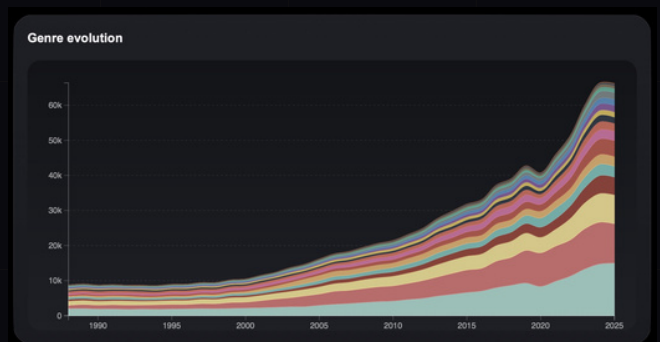
The Genre Evolution visualization shows the transition from our first hand-drawn sketch to the final deployed website.

Initially, we designed a simple stacked area chart to represent how genres evolved over time, while already imagining interactions like hover effects and filtering. This early sketch helped us define both the visual structure and the storytelling role of the graph within the website.

In the final implementation, the visualization was transformed into an interactive D3.js chart integrated into the cinematic narrative of the website.

When users hover over the graph, they can see the exact number of movies produced for a specific genre at a given time, making the data more precise and engaging.

First visualization gives a global image, while this one complements it, since users can now compare the global evolution of movie production with the evolution of individual genres.



## Popularity vs Quality heatmap

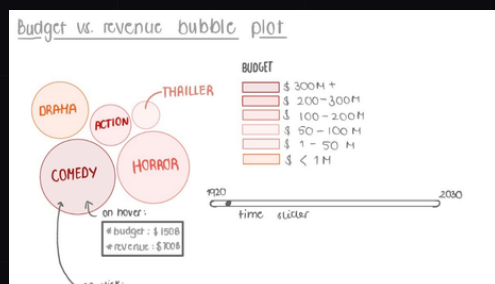


The prototype began as a hand-drawn heatmap concept, using genres, rating levels, and a color scale to show how popularity varies with quality.

In the final implementation, this was transformed into a clean digital heatmap with rounded cells, a dark cinematic background, clearer labels, rating numbers, star indicators, and an "All time" filter. The result keeps the same analytical idea but makes it more polished, readable, and consistent with the overall movie dashboard style.



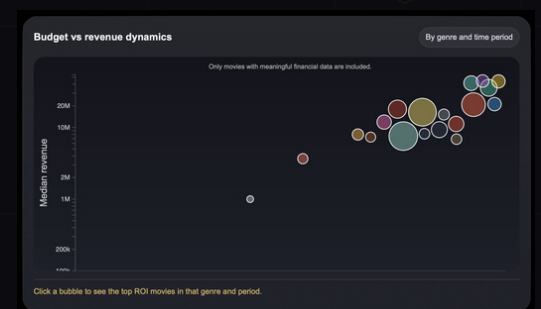
## Budget vs Revenue bubble plot



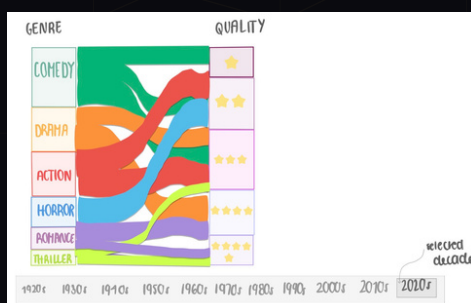
The visualization started with a hand-drawn prototype that defined the main interactions and layout of the bubble plot.

The goal was to represent movie genres through bubbles, with hover interactions showing information such as budget and revenue, and mouse clicks displaying more detailed movie data.

During implementation, these ideas were transformed into an interactive D3.js visualization with smooth animations, tooltips, and responsive mouse interactions, creating a more immersive and user-friendly experience while staying faithful to the original design.

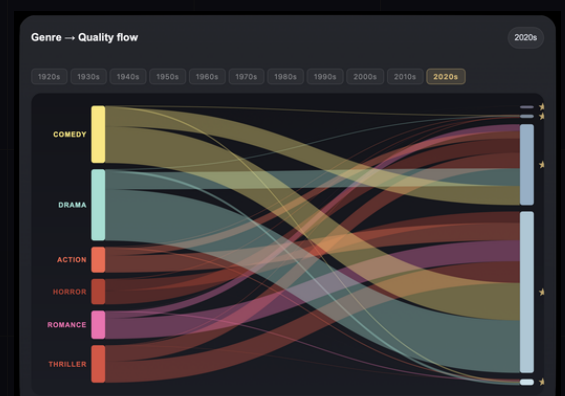


## Quality of Movies Through Time



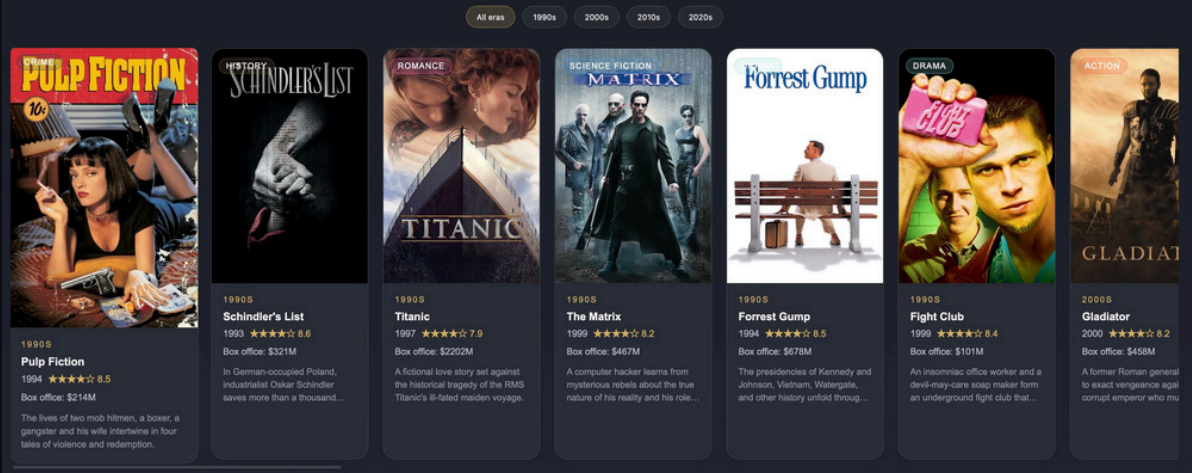
The prototype started as a simple hand-drawn concept showing how movie genres connect to different quality ratings over time.

In the implementation, this idea was transformed into a polished interactive-style Sankey visualization with a dark cinematic design, clearer genre labels, decade filters, smooth colored flows, and star-based quality levels. We added a video animation showing evolution over time.





# Representative Movies Panel



This section was designed to connect the data visualizations with concrete and recognizable movie examples. Instead of only presenting abstract statistics, we wanted users to visually explore iconic films from different eras through an interactive and scrollable gallery.

Users can horizontally scroll through movie cards, each displaying the film poster, genre, rating, release year, box office revenue, and a short overview. This makes the experience more immersive and engaging, while also reinforcing the cinematic atmosphere of the website. Additionally, each card provides direct access to the movie's IMDb page, allowing users to continue exploring the films beyond the visualization itself and creating a bridge between the data and real-world cinema culture.

## Your Cinema Era: Interactive Personalizer

While the other visualizations present cinema as a collective phenomenon, this section makes it personal. The era personalizer invites users to anchor themselves in the data by selecting any year between 1988 and 2024 and instantly seeing what cinema looked like at that moment in time.

A single slider controls four live statistics: the number of films released that year, the dominant narrative genre of the decade, the average audience rating, and the decade's highest-grossing blockbuster. Below the cards, a sparkline of the full production timeline keeps the selected year in context, with a dashed marker tracking the user's position across the curve.

The design decision behind this section was deliberate: data becomes meaningful when it connects to lived experience. Knowing that 13,485 films were released in 2004 is abstract: knowing that was the year you were born, that Avatar defined your decade financially, and that the average rating was 5.84 out of 10 makes the same numbers feel real. This section transforms the user from observer to participant.



# CHALLENGES

## Issues we faced

1. **Corrupted financial data:** 93.5% of budget values and 97.7% of revenue figures were recorded as zero in the raw dataset: not truly zero, but simply unknown. This silently distorted every financial analysis. ROI calculations produced infinite or meaningless values whenever budget was zero.
2. **Movie poster retrieval:** The representative films panel required real movie posters. The TMDb image CDN blocks all server-side HTTP requests, returning 403 errors regardless of headers or authentication tokens. The browser console script approach also failed because the hardcoded Bearer token had expired.
3. **The B&W → color visual arc:** Implementing a genuine color progression across the page proved surprisingly difficult. Multiple approaches failed: a single body gradient was overridden by opaque section backgrounds; background-attachment: fixed showed the same gradient slice in every viewport; per-section solid colors created harsh visible edges between sections; and per-section gradients that blended into each other produced colors that were too saturated and jarring. Meanwhile, the CSS filter: saturate() on sections correctly desaturated chart content, but the background itself remained a uniform blue-gray: undermining the narrative visually even when the charts were correctly desaturated.

## Solutions

1. **Corrupted financial data:** We replaced all zero values with null before aggregation and filtered them out of financial calculations entirely. The finance.json only includes films with meaningful budget AND revenue data. For the era personalizer's blockbuster lookup, we added an explicit revenue > 0 guard in the JavaScript.
2. **Movie poster retrieval:** We replaced the 7 broken films with 7 different landmark films whose TMDb poster paths we could confirm were correct from prior working examples. We also implemented a graceful onerror fallback in JavaScript: if any poster fails to load, the card displays a colored gradient with the film title instead: maintaining visual consistency even when images are unavailable. For future-proofing, we wrote a fetch\_poster\_paths.py script that retrieves correct paths from the TMDb API given a valid API key.
3. **The B&W → color visual arc:** After iterating through several approaches, we reverted to the original body gradient (black → dark blue-gray) combined with the original CSS saturate() filter scheme on each section. This proved to be the most stable and visually coherent solution: the saturation filters effectively desaturate content in early sections and gradually restore color as the user scrolls, while the background gradient provides a consistent atmospheric base without introducing harsh color jumps between sections.

# CONTRIBUTIONS

| TEAM MEMBER              | CONTRIBUTIONS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rihab Belmekki           | Popularity vs Quality Heatmap (TMDB data pipeline, log-normalised popularity aggregation, hexagonal grid renderer, decade filter, per-row hover dimming, tooltip system) · Genre-Quality Sankey Chart (movie count aggregation by genre×tier×decade, cubic bezier ribbon layout, left/right node sizing, stacked flow geometry, b&w→colour saturation transition for pre-1960 decades, play/stop animation with adjustable speed, decade timeline strip, ribbon hover isolation) · Milestone 2 sketches (hand-drawn wireframes for Popularity vs Quality Heatmap, Budget vs Revenue Bubble Plot, and Quality of Movies Through Time) |
| Ghalia Bennani           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Kenza Oudrhiri Benaddach | Representative films panel (movies.json curation, card UI, decade filter tabs, IMDB links, poster fallback system) · Era personalizer (slider interaction, animated stat cards, live JSON lookups, sparkline chart, Documentary exclusion logic) · Cinematic image integration (hero background, film countdown loader, Kodak strip divider, vertical filmstrip decoration, "The End" closing card) · Key findings section (data-backed insights for all 4 finding cards) · Process book: challenges page + era personalizer section                                                                                                 |
| Adam Mellouk             | Data exploration + data cleaning + data exportation into usable JSON. Developed the HTML website code + design the early working version of the viz for milestone 2. Reworked the site for milestone 3. Implemented movie production / genre evolution. Adapted genre buttons to influence the whole website. Process book : designed the template and overall layout + wrote introduction page / first sections.                                                                                                                                                                                                                    |